

Sainath's 6-Month Backend SWE + AI Career Roadmap

1. Career Positioning

Target Position: Backend Software Engineer with AI Integration Expertise. Your differentiation is not being another Java developer or another AI enthusiast. You should position yourself as someone who can build scalable backend systems and integrate AI capabilities into products.

2. End Goal After 6 Months

- 300+ DSA problems solved
- Strong Java fundamentals
- SQL and Database Design proficiency
- Spring Boot expertise
- Docker and JWT knowledge
- 3 strong projects
- Optimized GitHub and LinkedIn profiles
- Interview-ready resume

3. Daily Study Schedule (8–10 Hours)

Morning (3 hrs): DSA Concepts + Problem Solving

Mid Noon (2–3 hrs): Backend Learning (Java/SQL/Spring Boot)

Evening (2–3 hrs): DSA Practice and Revision

Night (2 hrs): Java Core, OOP, Collections, Mini Projects

4. Month 1 Roadmap

Java Basics, OOP, Arrays, Strings, Time Complexity, SQL Basics. Goal: Become comfortable writing Java programs without tutorials.

5. Month 2 Roadmap

Linked Lists, Stacks, Queues, HashMaps, Collections Framework, SQL Joins and Indexes. Goal: Solve medium-level DSA problems confidently.

6. Month 3 Roadmap

Trees, Heaps, Recursion, Spring Boot Basics, REST APIs, JPA/Hibernate. Start building a Task Manager API.

7. Month 4 Roadmap

Graphs, Binary Search, JWT Authentication, Spring Security. Begin E-Commerce Backend Project.

8. Month 5 Roadmap

Dynamic Programming, Docker, Redis, Deployment. Upgrade E-Commerce Backend and begin HealMate AI project.

9. Month 6 Roadmap

System Design Basics, Mock Interviews, Resume Preparation, Job Applications. Complete Chat Application and polish portfolio.

10. DSA Roadmap

Arrays → Strings → Hashing → Two Pointers → Sliding Window → Linked Lists → Stack → Queue → Trees → Heap → Graphs → DP. Target: 300+ problems.

11. Backend Roadmap

Java → SQL → Spring Boot → REST APIs → JPA → JWT → Docker → Redis → Microservices.

12. Project Strategy

Project 1: HealMate (AI Healthcare Assistant using LLMs and RAG)

Project 2: E-Commerce Backend (Spring Boot, JWT, Docker)

Project 3: Real-Time Chat App (WebSockets, Spring Boot)

13. GitHub Strategy

Every repository should contain: README, Architecture Diagram, Setup Instructions, Screenshots, API Documentation, and Deployment Links.

14. LinkedIn Strategy

Post project updates, architecture decisions, lessons learned, and technical insights. Avoid generic motivational posts.

15. Resume Strategy

Highlight measurable achievements, technologies used, and impact created through projects. Focus on backend engineering and AI integration.

16. Interview Preparation

Practice coding rounds, core CS subjects (DBMS, OS, CN), behavioral questions, and project explanations.

17. Competitive Advantage

Most candidates know Java or AI. Your advantage comes from combining strong backend engineering with practical AI integration. Build products, not just tutorials.

18. Final Priority Order

1. Java Fundamentals
2. DSA
3. SQL
4. Spring Boot
5. E-Commerce Backend
6. HealMate AI
7. Chat Application
8. Docker
9. GitHub Portfolio
10. LinkedIn Optimization
11. Job Applications